MECHANICAL ENGINEERING · UC RIVERSIDE



Kian Zarazvand

I design suspension, mechanisms and cryostat hardware — then build them in house and find out what breaks.

I'm a mechanical engineering student at the University of California, Riverside, expecting to graduate in 2029. I started in FIRST Robotics in 2022 on Team 6560 Charging Champions, ran a 30-student team as captain, and now mentor two programs while working on suspension for UCR Formula SAE and cryogenic hardware for the Dark Matter & Neutrino Lab. Most of what I know came from iterating on a part until it stopped breaking.

DEGREE	BS Mechanical Engineering, UC Riverside · expected June 2029
GPA	3.9
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Experience

May 2026 — present	**Mechanical Engineer** Dark Matter & Neutrino Lab, UC Riverside
Sep 2026 — present	**Undergraduate Researcher, simulation** RAMS Lab, UC Riverside
Oct 2025 — present	**Suspension Intern → Associate Engineer** UCR Formula SAE · Highlander Racing
Jun 2025 — present	**Project Manager & Design Mentor** FIRST Robotics
May 2024 — May 2025	**Team Captain** FIRST Robotics, Team 6560
Aug 2023 — Apr 2025	**CAD Lead** FIRST Robotics, Team 6560
Jun 2023 — Sep 2025	**Design Engineer, elevator mechanisms** FIRST Robotics, Team 6560
2022 & 2023	**Assistant Stage Manager** University High School Technical Theatre

Skills & tools

- CAD and simulation: SolidWorks (design and simulation), Fusion 360 (CAD/CAM), Onshape, FEA, skeleton modeling, GD&T, Design for Manufacturability.
- Manufacturing: CNC machining, laser cutting, 3D printing, sheet metal, carbon fiber layup, billet machining, welded and epoxy-bonded joints.
- Analysis: MATLAB optimization, multivariable calculus, linear algebra, stress calculations, simulation-testing code.
- Project management, part procurement, technical mentorship, sponsorship and budgeting.
- Power transmission: belts, #25 chain, gear racks, bevel gears, custom gearboxes, turntables.
- Sensing and control integration: hall-effect sensors, relative encoders.

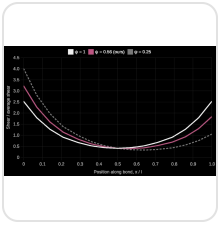
Education

- BS Mechanical Engineering, University of California, Riverside — 3.9 GPA, expected June 2029.
- Coursework: MATLAB, multivariable calculus, linear algebra, GD&T, 3D CAD design.
- University High School, Irvine — graduated June 2025 with a 4.15 GPA.

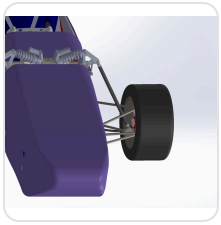
UCR FORMULA SAE · HIGHLANDER RACING

Formula SAE suspension

01 ANALYSIS



02 CAD



Carbon fiber pushrods

Associate Engineer, Suspension · Highlander Racing FSAE · May 2026 — present · critical design review, pre-build

Pushrods only carry axial load, which makes them the right first parts to move from steel to bonded carbon tube, and the testbed for the bonding process the team needs before a full carbon suspension. I researched the bonded-joint theory, worked out the Volkersen shear-stress distribution and the critical bondline length, then sized the front and rear tubes and their flanged 7075 inserts, ran the hand calculations and FEA, and presented the design at critical design review.

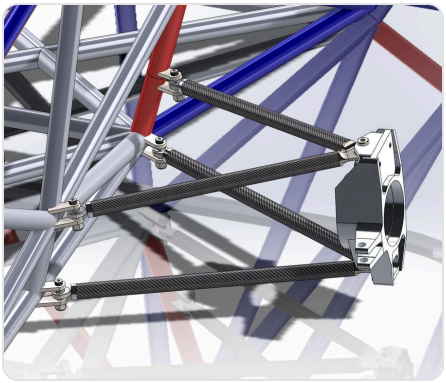
DESIGN & ANALYSIS

- Front: 0.625 × 0.750 in DragonPlate roll-wrapped twill tube at 4005 N peak load. Rear: 0.500 × 0.625 in at 5387 N. Both have a 1/16 in wall, with 7075-T6 inserts (503 MPa yield) and Loctite EA E-120HP epoxy.
- Roll-wrapped over pultruded: the 0/90 woven layers resist impact, splitting and torsion while the 0° unidirectional plies carry the axial load. Conservative laminate modulus of 11 msi from a rule-of-mixtures hand calculation.
- Researched adhesive-bond theory and worked through the modified Volkersen shear-lag solution for tubular lap joints (Jaeger & Wartzack, 2022): shear peaks at both bond ends at 3.2–3.3× the average, and an equal-stiffness ($\psi = 1$) insert and tube would equalize those peaks.
- Calculated the critical bondline length $l_{crit} = 5 \cdot \sqrt{(E_1 A_1 t / ((1+\psi) G \cdot w))}$: 27.6 mm front, 27.4 mm rear. Past it the middle of the bond carries almost no load, so extra length only adds mass.
- Moved the threads out of the bond zone so the insert keeps its full 1/16 in wall there ($\psi = 0.56$ front, 0.53 rear), and bounded the flange's share of the load at 20–50% with FEA.
- Buckling by hand: Euler on the front (14.5 kN critical load, FOS 3.6) and Johnson on the rear (25.9 kN, FOS 4.8), with pin-ended rod ends.
- Aurora XAM-4 / XAB-4 rod ends on 5/16-24 threads, left- and right-hand, make each pushrod a length-adjustable turnbuckle locked with jam nuts. Helicoils make worn threads replaceable.

BUILD, VALIDATION & RESULTS

- Four pushrods save 314 g per car against 4130 steel: front 70.5 g (−99 g, 59% lighter), rear 34.4 g (−58 g, 63% lighter), lowering unsprung mass.
- Bond FOS 4.8 front and 2.8 rear, assuming the flange carries no load; a 20–50% flange share would raise the rear to 4.0–5.3. Required FOS: 3 on peak bond shear, 2 on buckling.
- FEA on the front insert stays under 67 MPa at a 50/50 flange/shear split (FOS above 7.5 against yield) and near 50 MPa peak at 80/20.
- No carbon touches aluminum: an epoxy film on every interface, with glass beads holding the bondline, also takes up the 7075 versus carbon thermal expansion mismatch.
- Tubes and carbon tooling cost about $472 for the full set, with the tooling reused across seasons.
- Open items called out in the review: laminate modulus and strength are working assumptions until coupons are tested, and the thermal mismatch at the 65 °C post-cure is not yet analyzed.
- Assembly and validation procedure written alongside: surface prep, helicoils before bonding, dial-indicator alignment, then Instron compression and bond coupons.

CAD



Full carbon suspension concept

Suspension Intern · January 2026 · concept study, not validated or built

My intern intro project: a first pass at a full carbon fiber suspension, with carbon tube arms joined by billet steel inserts and tabs. It was a CAD and hand-calculation concept — ambitious, but never validated or built. The current pushrod project is the realistic step toward it: prove bonded carbon parts properly first.

DESIGN ELEMENTS

- Skeleton modeling, FEA and stress calculations with Design for Manufacturability in mind.
- Billet steel inserts bonded into carbon tubes, with steel tabs to the chassis.
- Ball joints, rod ends and welded joints for a configurable, adjustable layout.

FUNCTIONALITY

- Concept only: not validated, tested or built.
- Showed that the bonded carbon-to-metal joint is the thing to prove first, which is what the pushrod project does.

UCR DARK MATTER & NEUTRINO LAB

Cryogenic research hardware

01 CAD



02 PROTOTYPE



Lumirror sheet housing

Mechanical Engineer · May 2026 – present

The housing has to keep a 1 m × 8 m reflective plastic sheet taut and located inside a cryostat while the whole assembly contracts from room temperature down to 70 K. The aluminum prototype is assembled and holding sheet; the final build is next.

DESIGN ELEMENTS

- CNC aluminum for the prototype, now assembled and tested with sheet.
- Laser-cut 304 stainless steel planned for the final build.
- Sheet metal tabs and clamping sheets to grip the film without tearing it.
- Fastening designed around rapid cryogenic thermal contraction.

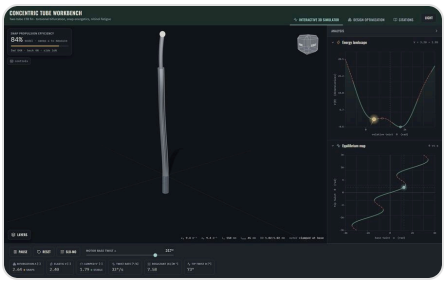
FUNCTIONALITY

- Tensions and houses a 1 m × 8 m sheet of Lumirror plastic.
- Withstands rapid thermal contraction from room temperature to 70 Kelvin.

UCR RAMS LAB · CONCENTRIC TUBE ROBOTICS

Snap-driven swimming robot

ANALYSIS



Concentric tube snap simulator

Undergraduate Researcher, RAMS Lab · UC Riverside · Sep 2026 — present

Medical concentric tube robots treat the snap of pre-curved tubes as a failure to avoid. The RAMS Lab uses it on purpose: a fin that stores torsional energy and releases it in a burst-and-coast swimming stroke. Since joining the group I have been responsible for the simulator, using it to re-evaluate whether our original fin plan was the most efficient and to explore other ways of harnessing the Nitinol snap to propel the robot.

Live tool: qwertz101.github.io/turtle-robot-sim/

DESIGN ELEMENTS

- Two-tube model built on the Gilbert–Hendrick–Webster elastic stability analysis: bifurcation parameter λ, snap onset at $\lambda_0 = \pi^2/4$, energy landscape and equilibrium map.
- Design optimization surface of physical snap energy E_snap = ΔE · k_t / L_c (mJ) across precurvature and overlap length, scored by propulsion efficiency and Nitinol fatigue strain.
- Free, fully constrained and planar outer-tube modes to compare how the snap energy splits between bending and torsion.
- Photoreal three.js view with physically consistent shadows, slow-motion snap replay and a SolidWorks-style view cube.

FUNCTIONALITY

- Traces the fin's motion path, colour-coded build-up vs snap, and resolves a net propulsion vector shown as a compass.
- Snap propulsion efficiency: the share of each snap that pushes along the net propulsion direction.
- Every quantity carries units and its source equation; a citations page covers every paper used.
- Deployed on GitHub Pages so the whole group can use it from a browser.

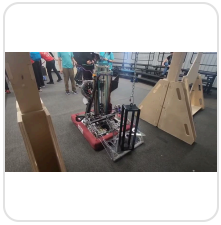
FIRST ROBOTICS · TEAM 6560 CHARGING CHAMPIONS

Competition robot mechanisms

01 CAD



02 IN COMPETITION



Climbing rotary mechanism

Team Captain & CAD Lead · February 2025

The end-game climb had to catch a swinging steel cage, then rotate it into the robot hard enough to lift 150 lb off the floor — and survive being slammed into the cage at speed, match after match.

DESIGN ELEMENTS

- Utilized 3/8" polycarbonate for robustness.
- Engineered a gusset-and-tube structure for simplicity.
- Created an impact-resistant funnel with polycarbonate plates.
- Used sheet metal bending to lower assembly complexity.

FUNCTIONALITY

- Funnels a steel "cage" into the mechanism.
- Lifts the 150 lb robot by rotating the cage into the robot.

01 CAD



02 PROTOTYPE



03 BUILT



Offset pivot rotary mechanism

Team Captain & CAD Lead · January 2025

The arm needed to pivot from the top of the elevator without its gearbox riding along for the trip. Offsetting the pivot through a large turntable and a chain run let the gearbox sit low while the arm stayed stiff.

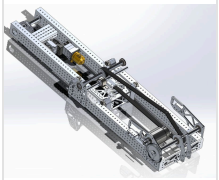
DESIGN ELEMENTS

- Used #25 chain for lightweight, high-torque power transmission.
- Simplified the design with an inline turnbuckle chain tensioner.
- Increased joint stiffness with a large-diameter turntable.
- Used an external relative encoder to account for gearbox backlash.
- Lowered the gearbox to lower the mechanism centre of gravity.

FUNCTIONALITY

- Offsets the pivot of an "arm" mechanism.
- Integrates with the "elevator" mechanism.

01 CAD



02 BUILT



2-stage linear motion + rotational gearbox

CAD Lead · January 2024

Packaging is the whole problem here: the wrist gearbox lives inside the elevator carriage, the belt is diverted around it, and a constant-force spring cancels the mechanism's own weight so the motor only moves the payload.

DESIGN ELEMENTS

- Used a bevel gear to optimize the gearbox packaging.
- Diverted the belt for packaging.
- Utilized a 10 lb constant-force spring to offset the mechanism weight.

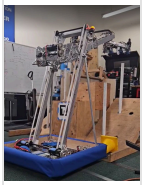
FUNCTIONALITY

- Extends a "wrist" mechanism to score game pieces.
- Houses the "wrist" gearbox in the mechanism carriage.

01 CAD



02 BUILT



03 IN COMPETITION



3-stage linear motion mechanism

Design Engineer · July 2023

The mechanism I have iterated on longest. Cascade rigging lets one motor drive three stages, and the belt tensioning was redesigned until it was continuous, easy to make and did not skip under load. Over twelve revisions across two years, the family of elevators that grew from this design reached 60 inches of travel in 0.4 seconds and a 300 lb lifting capacity.

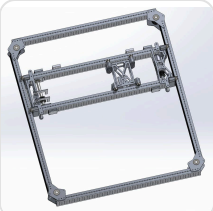
DESIGN ELEMENTS

- Utilized cascade rigging to extend multiple stages with one motor.
- Customized the belt tensioning to be continuous and easily manufacturable.
- Pocketed CNC plates to meet weight limits.
- Iterated over 7 times to perfect the design.
- Later revisions added A-frames, custom gearboxes and hall-effect sensors for reliability and control.

FUNCTIONALITY

- Moves items double the length of the mechanism.
- Transports game pieces to score at a distant location.
- Design family rated for 60 in of travel in 0.4 s and 300 lb lifting capacity.

01 CAD



02 BUILT



3D printer training task

New member training project · June 2023

A two-axis gantry in the style of a 3D printer, built as a training exercise that became the team’s testbed for belted linear motion.

DESIGN ELEMENTS

- Protected the powertrain by belting inside the tubestock.
- Tensioned the belt to minimize backlash.
- Fortified the structure with corner gussets.
- Used 6061 aluminum tubestock for the structure.

FUNCTIONALITY

- Provides research and development for future linear motion mechanisms.
- X and Y axis degrees of freedom.

FIRST ROBOTICS · ELEVATOR LINEAGE · 2023 → 2025

One mechanism, re-optimized every season

I have been the team's elevator specialist since 2023. Each season the game asked for something different — more reach, less weight, tighter packaging, more load — so the same linear-motion mechanism was re-optimized around new requirements. Over 12 revisions in two years, the family reached 60 in of travel in 0.4 s and a 300 lb lifting capacity.

	MULTI-STAGE	LIGHTWEIGHT	COMPACT	SIMPLE	HEAVY-DUTY	FAST
V0 Gantry trainer Jun 2023				●		
V1 3-stage cascade elevator Jul 2023	●				●	
V2 2-stage elevator + rotational gearbox Jan 2024			●	●		
V3 Elevator with offset arm pivot Jan 2025			●			●

BUILT



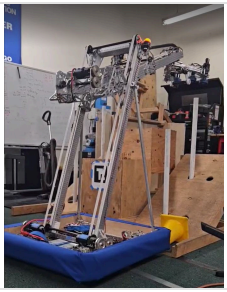
V0 · Jun 2023

Gantry trainer

Requirement: Learn belted linear motion before building it on a robot.

- Belts run inside 6061 tubestock to protect the powertrain.
- Belt tensioned to minimize backlash; corner gussets for stiffness.
- Became the testbed for every elevator after it.

BUILT



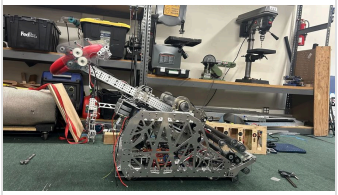
V1 · Jul 2023

3-stage cascade elevator

Requirement: Score at a distant location: travel double the mechanism length.

- Cascade rigging drives three stages from one motor.
- Continuous, easily manufactured belt tensioning.
- Pocketed CNC plates to meet the weight limit; 7+ iterations.

BUILT



V2 · Jan 2024

2-stage elevator + rotational gearbox

Requirement: Carry a powered wrist without growing the robot.

- Dropped to two stages; the wrist gearbox lives inside the carriage.
- Bevel gear and a diverted belt path for packaging.
- 10 lb constant-force spring cancels the mechanism weight.

BUILT



V3 · Jan 2025

Elevator with offset arm pivot

Requirement: Hold a long pivoting arm at full height without flex.

- Large-diameter turntable stiffens the arm joint.
- #25 chain with an inline turnbuckle tensioner.
- Gearbox lowered to drop the mechanism centre of gravity.

LEADERSHIP

Running the team



Team captain

Team Captain, Team 6560 · Irvine, CA · May 2024 — May 2025

Captaining meant owning the whole pipeline — not just the robot, but the people, money and schedule that produce one on a six-week clock.

DESIGN ELEMENTS

- Led a 30-student team in designing, building, wiring and coding a competition robot in two months.
- Managed recruitment, reimbursements, sponsorships and robot integration across subteams.
- Handled project management across the mechanical, electrical and software groups.

FUNCTIONALITY

- Competed in five yearly competitions with the resulting machine.



CAD lead

CAD Lead, Team 6560 · Irvine, CA · August 2023 — April 2025

This was where the team's design practice got built: a shared library, a file system people could actually navigate, and six designers trained to the same standard.

DESIGN ELEMENTS

- Managed a team of 6 designing 400+ component robots with CNC, laser cutter and 3D printer in-house manufacturing and prototyping.
- Led R&D efforts to expand the design library and organized the team hierarchy for better productivity across skill levels.
- Created a Google Drive file management system and managed part procurement.

FUNCTIONALITY

- Trained new members to SolidWorks proficiency.
- Two full competition robots shipped on schedule.



Design mentor & project manager

Irvine High School Robotics & Charging Champions Robotics · June 2025 — present

Two strands of the same skill: getting a group of people to deliver a working physical system on a fixed date.

DESIGN ELEMENTS

- Provide design mentorship and CAD training for Irvine High School Robotics and Charging Champions Robotics.
- Project management for a controls team working in automation, vision, object detection and general robot programming.
- Implemented concurrent design and manufacturing, and standardized simulation-testing code.

FUNCTIONALITY

- As Assistant Stage Manager for University High School Technical Theatre, managed a crew of 10 through the technical set-up and planning of lighting, scenery and audio for productions (2022, 2023).